Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/31/2025
PI Name:	Riaz, Amir Ph.D.		
Project Title:	Pool Boiling Heat Transfer Mechanisms in Low Gravity: Numerical Experiments of MABE and NPBX Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ariaz@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-0707
Organization Name:	University of Maryland		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	127 Glenn L Martin Hall		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017/2019 Physical Sciences NNH17ZTT001N-17PSI-F: Use of the NASA Physical Sciences Informatics System – Appendix F
Start Date:	12/01/2020	**End Date:**	11/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2025 per NSSC information (Ed., 12/4/24) NOTE: End date changed to 11/30/2024 per PI (Ed., 11/27/23) NOTE: End date changed to 11/30/2023 per NSSC information (Ed., 11/22/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balaras, Ilias Ph.D. (George Washington University)		
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Task Description:	The key challenge in developing a proper understanding of nucleate boiling in thermal systems for space environments is the accurate characterization of bubble dynamics and associated heat transfer mechanisms over a wide range of wall superheat and liquid subcooling conditions. To address this challenge, we are proposing well-characterized numerical experiments using high fidelity Direct Numerical Simulation for investigating the dynamics of nucleate boiling across a range of gravity conditions. Our numerical experiments will recreate the conditions of the Microheater Array Boiling Experiment (MABE) and Nucleate Pool Boiling eXperiment (NPBX) that were flown on the International Space Station (ISS) in 2011 within the Boiling Experiment Facility (BXF). MABE provided local heat transfer and visualization of pool boiling heat transfer behavior in a microgravity environment. The focus of NPBX was the investigation of how bubbles grow and merge in microgravity. The main finding was that heat fluxes for steady state nucleate boiling were found to be substantially lower than what was expected on the basis of low gravity data obtained on Earth. The numerical solver that we will use to simulate MABE conditions has been recently validated by comparing the heat flux from our fully 3-D simulations to MABE data for a representative sample of wall superheat and liquid subcooling conditions. Details about our numerical methods and validation case studies have been recently published (Akash Dhruv, Elias Balaras, Amir Riaz, and Jungho Kim. A formulation for high-fidelity simulations of pool boiling in low gravity. International Journal of Multiphase Flow, 120:103099, 2019). These case studies demonstrate that our solver is able to resolve fully the complex interactions of liquid and vapor phases involving phase transition, interphase mass flux, and contact line dynamics. Using the data from the simulations, we have been able to connect the dynamics of bubble nucleation and growth to the corresponding heat flux measurements in MABE data. Such detailed characterizations of the flow field have already started to shape and improve our understanding of the underlying bubble dynamics. In the proposed research, we aim to systematically probe the MABE database now at a deeper level to extract fundamental information on: (i) how is heat flux partitioned into contributions arising from nucleate boiling and natural convection? (ii) what is the role of contact angle in affecting the force of wall adhesion? and (iii) how much heat transfer occurs through microlayers in low gravity settings? These questions will be addressed to qualitatively different regimes of bubble dynamics under various conditions of gravity, wall superheat, and liquid subcooling. The answers to these questions, investigated on the basis of a validated solver using MABE data, would extend the general understanding of pool boiling in low gravity settings. The use of the MABE database is instrumental to this endeavor. It will be used to provide the reference dataset that connects heat flux, and in some cases even visualization of nucleate boiling, to different levels of gravity, wall super heat, and subcooling. This will allow a unique opportunity for a quantitative assessment of potential improvements in the design of thermal systems in space applications. The proposed research is expected to complement NASA's ongoing efforts, in addition to advancing the science of heat flux partitioning, contact line dynamics, and the role of microlayers for nucleate boiling in low gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Electronic devices require effective and efficient means of thermal management to achieve performance targets. The use of latent heat removal via the process of phase change, such as nucleate and pool boiling, is an extremely effective method of thermal management. This method relies primarily on the dominance of buoyancy over capillary forces to maintain high heat flux at the heated surface. The proposed research will study the influence of major parameters on the behavior of boiling heat transfer. The resulting data and analysis will help improve the thermal management of electronic devices, enabling the achievement of higher performance goals.
Task Progress:	Numerical simulations were carried out to investigate the effect of subcooling on pool boiling and the effect of biphilic surfaces on pooling boiling in Earth and low gravity. Two manuscripts have resulted from these studies. One titled, “Numerical simulation of pool boiling on biphilic surfaces”, has appeared in the International Journal of Heat and Mass Transfer 239 (2025) 126531 and the second manuscript titled, “Direct numerical simulations of subcooled pool boiling: Studies of heat transfer and bubble dynamics” has been published in the Physics of Fluids 37, 023393 (2025); doi: 10.1063/5.0254660. The graduate students presented their work at the American Society of Gravitational and Space Research and the American Physical Society, Division of Fluid Dynamics. One graduate student graduated in 2025 and second student is scheduled to graduate in 2026. Task 1. Study of the dynamics of vorticity associated with subcooled nucleate pool boiling Numerical experiments were caried out using FC-72 as a working fluid with the objective of understanding the dynamics of vorticity associated with the bubbles in terms of the effect on subcooling that led to a minimum and an inflection point in heat flux. The results indicate that at low subcooling before the heat flux reaches the inflection point, the bubble dynamics are governed by growth, merging, and departure events which give rise to predominantly ring vortices structures. In the case of higher subcooling, beyond the inflection point, the dynamics of the bubbles are characterized by the absence of departures, a reduction of the bubble size and growth rate, as well as qualitatively different form of merging and coalescence which gives rise to predominantly hair-pin vortex structures. More details are provided below and, in the article, Physics of Fluids 37, 023393 (2025). Task 2. Study of turbulence statistics for pool boiling on biphilic surfaces. Analysis of numerical experiments of pool boiling were carried out to help identify the key physical mechanisms responsible for enhancing heat transfer on biphilic surfaces. Using a uniform array of square hydrophobic islands on a hydrophilic substrate, the thermal performance of biphilic surfaces is characterized as a function of the ratio of hydrophobic and hydrophilic surface areas. The heater is isothermal and all other parameters are held fixed. Overall, the results indicate that, similar to experimental observations, a peak in the heat transfer rate occurs for an intermediate area ratio. This study finds, as reported in the Annual Report 2023, that the nucleated bubbles are attracted preferentially to the hydrophobic islands where they become confined. When these bubbles grow, they merge to form large vapor structures which are anchored to the heater surface at multiple islands, as reported in the Annual Report 2023. As a result, a thin evaporating liquid layer is trapped under the vapor bridge, between the wall and the vapor–liquid interface. The steep temperature gradient within this liquid layer emerges as the characteristic mechanism associated with the high heat flux on biphilic surfaces. Bubble pinning on the hydrophobic islands, heat flux partitioning associated with bridge formation and the mechanism of thin layer evaporation were described in the Annual report 2024. This Annual Report of 2025, describes the use of turbulent heat flux for the analysis of the impact of island size on bridge formation. More

	details are provided below and, in the article, International Journal of Heat and Mass Transfer 239 (2025) 126531.
Bibliography Type:	Description: (Last Updated: 04/23/2026)
Abstracts for Journals and Proceedings	Youssoufi S, Balaras E, Riaz A. "Vortex and turbulence statistics in nucleate pool boiling." 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21), Busan, Korea, August 31-September 5, 2025. Abstracts. 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21), Busan, Korea, August 31-September 5, 2025. , Sep-2025
Articles in Peer-reviewed Journals	Vachhani S, Riaz A, Balaras E. "Numerical simulation of pool boiling on biphilic surfaces." International Journal of Heat and Mass Transfer. 2025 Apr 1;239:126531. https://doi.org/10.1016/j.ijheatmasstransfer.2024.126531 , Apr-2025
Articles in Peer-reviewed Journals	Youssoufi S, Riaz A, Balaras E. "Direct numerical simulations of subcooled pool boiling: Studies of heat transfer and bubble dynamics." Physics of Fluids. 2025 Feb 1;37(2). https://doi.org/10.1063/5.0254660 , Feb-2025